

# Line Segments Trace II

Time limit: 1.00 s

Memory limit: 512 MB

There are  $n$  line segments whose endpoints have integer coordinates. Each x-coordinate is between 0 and  $m$ . The slope of each segment is an integer.

For each x-coordinate  $0, 1, \dots, m$ , find the maximum point in any line segment. If there is no segment at some point, the maximum is  $-1$ .

## Input

The first line has two integers  $n$  and  $m$ : the number of line segments and the maximum x-coordinate.

The next  $n$  lines describe the line segments. Each line has four integers  $x_1, y_1, x_2$  and  $y_2$ : there is a line segment between points  $(x_1, y_1)$  and  $(x_2, y_2)$ .

## Output

Print  $m + 1$  integers: the maximum points for  $x = 0, 1, \dots, m$ .

## Constraints

- $1 \leq n, m \leq 10^5$
- $0 \leq x_1 < x_2 \leq m$
- $0 \leq y_1, y_2 \leq 10^9$

## Example

| Input                                           | Output       |
|-------------------------------------------------|--------------|
| 4 5<br>1 1 3 3<br>1 2 4 2<br>2 4 5 7<br>2 8 5 2 | -1 2 8 6 6 7 |